

AZURE CLOUD SECURITY MONITORING LAB

Implementation & Technical Evidence Report

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1. Project Overview

This report documents the design, deployment, and validation of a cloud-based security monitoring solution built on Microsoft Azure. The lab simulates a real-world Security Operations Centre (SOC) workflow: provisioning a hardened Linux virtual machine, restricting network access using the principle of least privilege, centralising log collection through Azure Monitor and Log Analytics, and using Kusto Query Language (KQL) to detect and alert on SSH authentication activity.

1.1 Objectives

- Deploy and configure an Ubuntu Linux virtual machine in Microsoft Azure to simulate a secure cloud environment.
- Configure Network Security Groups (NSGs) and SSH key-based authentication to secure remote access using least privilege principles.
- Configure Microsoft Entra ID and Azure Monitor to centralise security monitoring and log collection.
- Create and associate a Data Collection Rule (DCR) to collect Linux Syslog data into an Azure Log Analytics Workspace.
- Write KQL queries to analyse SSH authentication events, session activity, and security logs.
- Create an Azure Monitor log alert to detect SSH login events and demonstrate automated security monitoring.

1.2 Technology Stack

Cloud Platform	Microsoft Azure
Monitoring & Logging	Azure Monitor, Log Analytics Workspace, Azure Monitor Agent (AMA)
Data Ingestion	Data Collection Rules (DCR) — Linux Syslog
Query Language	Kusto Query Language (KQL)
Identity	Microsoft Entra ID
Operating System	Ubuntu Server 24.04 LTS
Network Security	Network Security Groups (NSGs), SSH key-based authentication

1.3 Environment Summary

Resource Group	CloudSecurityLab-RG
Virtual Machine	CloudSec-Linux-VM
Region	East US 2
Admin Username	azureuser
Authentication Method	SSH public key

2. Phase 1 — Environment Provisioning

The foundation of the lab began with creating a dedicated resource group to logically contain and manage all lab assets, followed by deployment of the Linux virtual machine that would act as the monitored host.

2.1 Resource Group Creation

Purpose

A dedicated Resource Group was created to logically organise and manage every Azure resource used throughout the lab, in line with Azure resource-management best practice.

Configuration

Resource Group	CloudSecurityLab-RG
Subscription	Azure subscription 1
Location	East US 2

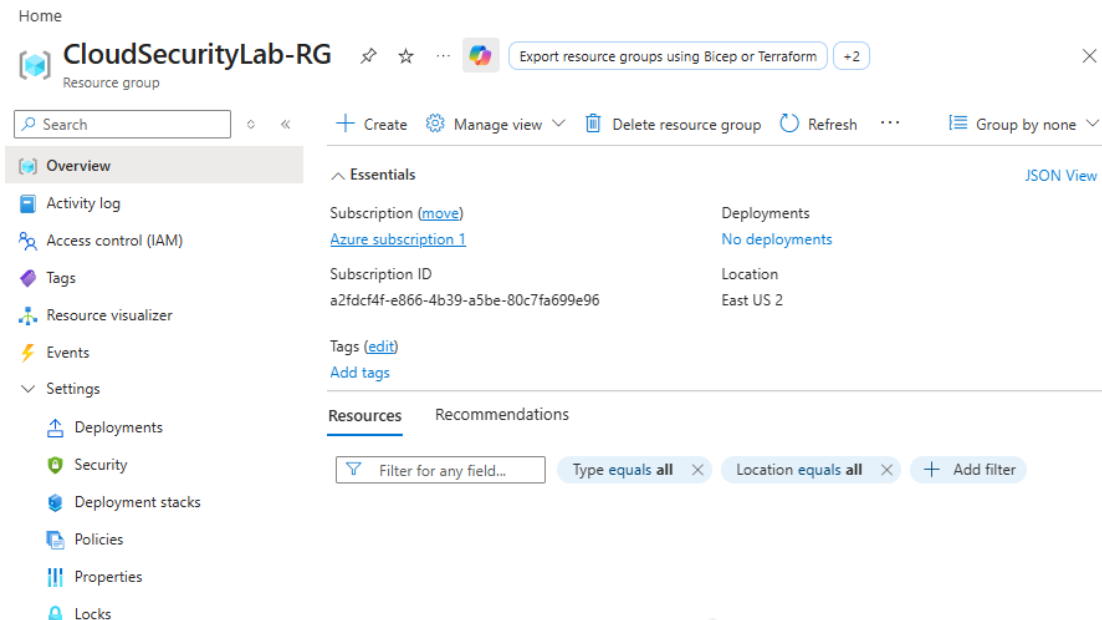


Figure 1: Resource group overview confirming successful creation of CloudSecurityLab-RG.

Observation

The resource group was created with no deployments initially present, confirming a clean environment ready for the virtual machine and monitoring resources to be provisioned into.

2.2 Virtual Machine Deployment

Purpose

An Ubuntu Linux virtual machine was deployed to serve as the target host for security monitoring — representing a typical Linux server that a SOC analyst would need to protect and observe.

Configuration

Deployment Name	CreateVirtualMachine-CloudSec-Linux-VM
-----------------	--

VM Name	CloudSec-Linux-VM
Image	Ubuntu Server 24.04 LTS
Size	Standard_D2s_v3
Resource Group	CloudSecurityLab-RG

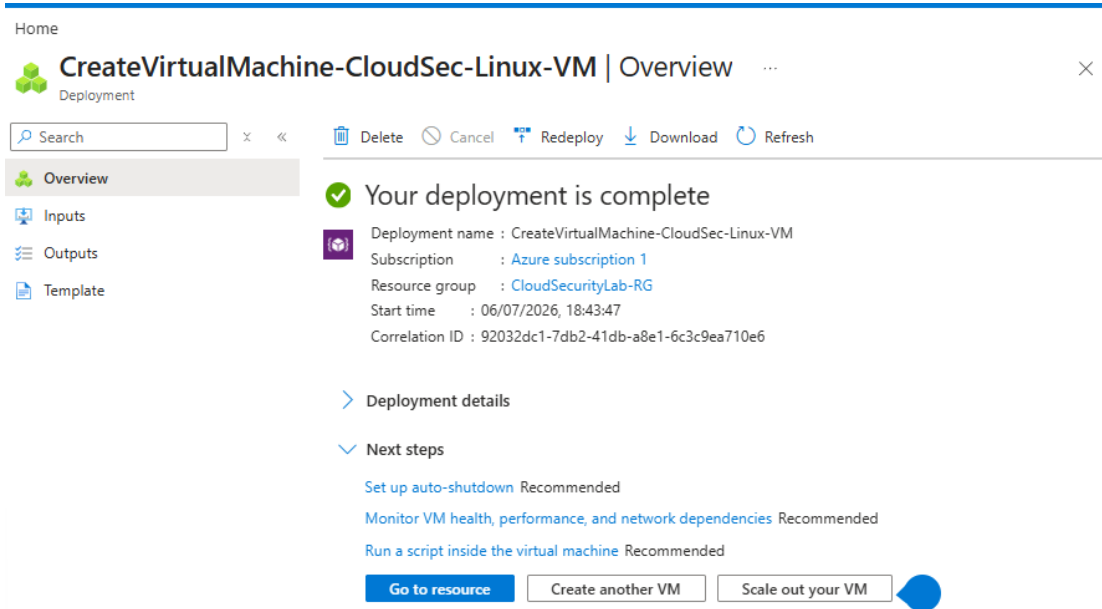


Figure 2: Deployment confirmation showing the virtual machine provisioned successfully.

Result

Deployment completed successfully, providing a live Linux host for subsequent network hardening and monitoring configuration.

3. Phase 2 — Network & Access Hardening

With the VM provisioned, remote access was locked down to reduce the attack surface, applying least-privilege principles to inbound network rules before verifying access and patching the operating system.

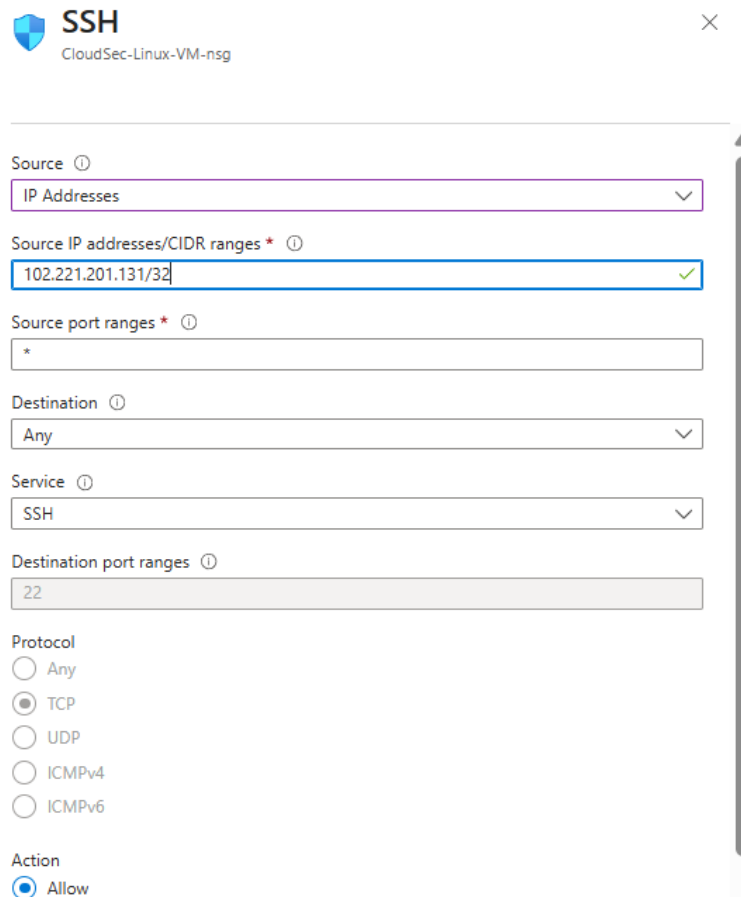
3.1 Network Security Group (NSG) — Least Privilege SSH Rule

Purpose

The default NSG inbound rule for SSH was restricted so that only a specific, known IP address could connect to the VM on port 22, rather than allowing access from any source (0.0.0.0/0). This directly applies the principle of least privilege to remote administrative access.

Configuration

NSG	CloudSec-Linux-VM-nsg
Source	IP Addresses — 102.221.201.131/32
Destination Port	22 (SSH)
Protocol	TCP
Action	Allow



SSH
CloudSec-Linux-VM-nsg

Source ⓘ
IP Addresses ▼

Source IP addresses/CIDR ranges * ⓘ
102.221.201.131/32 ✓

Source port ranges * ⓘ
*

Destination ⓘ
Any ▼

Service ⓘ
SSH ▼

Destination port ranges ⓘ
22

Protocol
☐ Any
☒ TCP
☐ UDP
☐ ICMPv4
☐ ICMPv6

Action
☒ Allow

Figure 3: NSG inbound security rule restricting SSH access to a single trusted source IP.

Observation

By scoping the source to a single /32 CIDR range instead of a broad range or "Any", the VM's exposed attack surface for brute-force or opportunistic SSH scanning is significantly reduced.

3.2 SSH Access Verification

Purpose

A successful SSH connection was established using key-based authentication to confirm that the NSG rule and SSH configuration were both working as intended.

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1018-azure x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Mon Jul  6 17:32:38 UTC 2026

System load:  0.08      Processes:           120
Usage of /:   5.8% of 28.02GB   Users logged in:    0
Memory usage: 3%          IPv4 address for eth0: 172.16.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@CloudSec-Linux-VM:~$
```

Figure 4: Successful SSH session into CloudSec-Linux-VM showing the Ubuntu 24.04.4 LTS welcome banner and system status.

Observation

The terminal banner confirms the host identity (CloudSec-Linux-VM), kernel version, and that the session authenticated correctly as the azureuser account — validating that key-based SSH authentication was functioning as configured.

3.3 System Patching

Purpose

Standard package updates were applied to the VM to ensure the operating system and installed utilities were current, reducing the risk of known vulnerabilities being present on the monitored host.

```

66 packages can be upgraded. Run 'apt list --upgradable' to see them.
azureuser@CloudSec-Linux-VM:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  fwupd
The following packages will be upgraded:
  ca-certificates cifs-utils curl gzip inetutils-telnet iproute2 kpartx
  libcurl3t64-gnutls libcurl4t64 libdrm-common libdrm2 libjcat1
  libncurses6 libncursesw6 libnghttp2-14 libnss-systemd libnss3
  libpam-systemd libperl5.38t64 libpython3.12-minimal libpython3.12-stdlib
  libpython3.12t64 libsqlite3-0 libssl3t64 libsystemd-shared libsystemd0
  libtinfo6 libudev1 libxml2 libxmb2 linux-cloud-tools-common
  linux-tools-common multipath-tools ncurses-base ncurses-bin ncurses-term
  openssl perl perl-base perl-modules-5.38 python3.12 python3.12-minimal
  rsync systemd systemd-dev systemd-resolved systemd-sysv tar telnet udev
  vim vim-common vim-runtime vim-tiny xxd
55 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
0 standard LTS security updates.
Need to get 48.4 MB of archives.
After this operation, 35.8 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 gzip amd64 1.12-1ubuntu3.2 [99.2 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 ncurses-bin amd64 6.4+20240113-1ubuntu2.1 [188 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libperl5.38t64 amd64 5.38.2-3.2ubuntu0.3 [4876 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 perl amd64 5.38.2-3.2ubuntu0.3 [231 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 perl-base amd64 5.38.2-3.2ubuntu0.3 [1827 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 perl-modules-5.38 all 5.38.2-3.2ubuntu0.3 [3110 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 tar amd64 1.35+dfsg-3ubuntu0.2 [256 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 ncurses-base all 6.4+20240113-1ubuntu2.1 [25.7 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 ncurses-term all 6.4+20240113-1ubuntu2.1 [275 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12t64 amd64 3.12.3-1ubuntu0.15 [2339 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libssl3t64 amd64 3.0.13-0ubuntu3.11 [1942 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3.12 amd64 3.12.3-1ubuntu0.15 [651 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12-stdlib amd64 3.12.3-1ubuntu0.15 [2071 kB]

```

Figure 5: Output of `sudo apt upgrade -y` showing package list evaluation and pending Ubuntu security updates.

Result

The system package index was refreshed, and outstanding updates were identified for installation, forming part of routine host hardening prior to enabling monitoring.

4. Phase 3 — Centralised Monitoring Foundation

To centralise security visibility, a Log Analytics Workspace was deployed as the destination for collected logs, forming the backbone of the monitoring pipeline that Azure Monitor and Microsoft Entra ID integrate with.

4.1 Log Analytics Workspace Deployment

Purpose

A Log Analytics Workspace was deployed to act as the central repository for log data ingested from the VM, enabling querying, analysis, and alerting through Azure Monitor.

Configuration

Deployment	Microsoft.LogAnalyticsOMS
Resource Group	CloudSecurityLab-RG
Subscription	Azure subscription 1

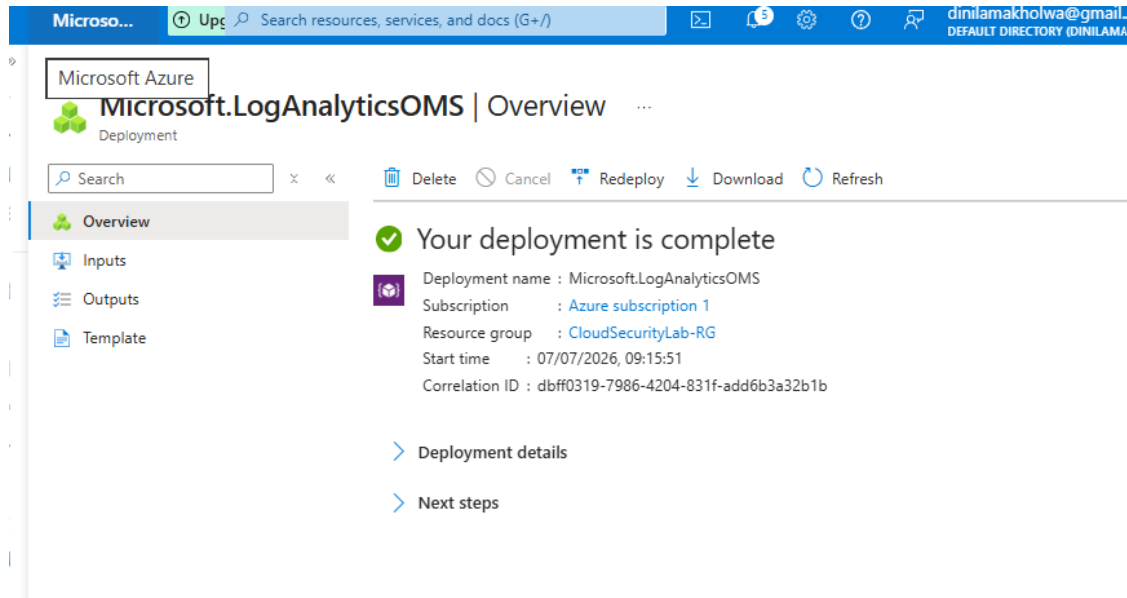


Figure 6: Successful deployment of the Log Analytics Workspace used to centralise log collection.

Result

The workspace deployed successfully and was ready to receive Syslog data once the Data Collection Rule was associated with the VM.

4.2 Pre-Monitoring Extension Check

Purpose

Before associating the Data Collection Rule, the VM's Extensions blade was reviewed to confirm the baseline state of the machine — specifically, that no monitoring agent extension had yet been installed.

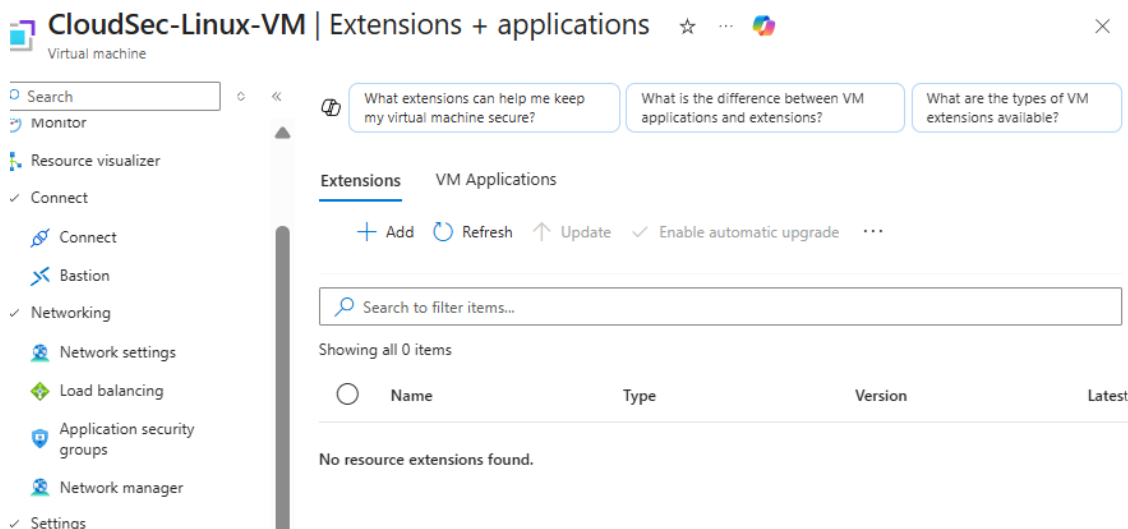


Figure 7: VM Extensions + Applications blade confirming no resource extensions were present prior to DCR association.

Observation

This baseline check confirms that the Azure Monitor Agent was not yet deployed, validating that log collection observed later in the lab was a direct result of the Data Collection Rule association rather than a pre-existing agent.

5. Phase 4 — Data Collection Rule (DCR) Configuration

Purpose

A Data Collection Rule was created and associated with the virtual machine to define exactly what log data (Linux Syslog) should be collected and where it should be routed — the Log Analytics Workspace deployed in Phase 3. Associating a DCR with a VM also triggers automatic installation of the Azure Monitor Agent.

Configuration

Deployment	DCR-Deployment-2026-07-07-197ed842-44f4-4faf-841f-60054aceff26
Data Source	Linux Syslog
Destination	Log Analytics Workspace (CloudSec-LogAnalytics)
Resource Group	CloudSecurityLab-RG

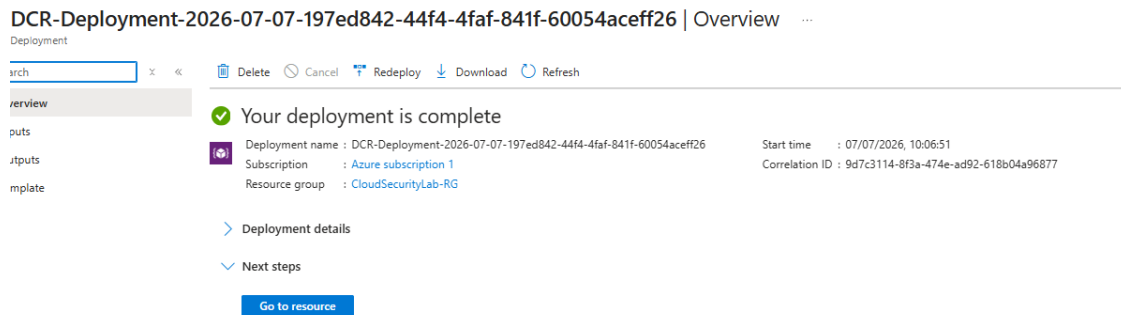


Figure 8: Successful Data Collection Rule deployment linking the VM's Syslog data to the Log Analytics Workspace.

Result

The DCR deployed successfully, establishing the pipeline required for Syslog events (including SSH authentication logs) to flow from the VM into Log Analytics for querying.

6. Phase 5 — KQL Log Analysis

With Syslog data flowing into the workspace, KQL was used to progressively narrow down and analyse SSH-related security events: authentication attempts, session lifecycle activity, and disconnect events.

6.1 Query 1 — SSH Authentication Events

Purpose

An initial broad query filtered the Syslog table for events generated by the sshd process, to confirm that SSH activity was being logged and to review the raw event structure.

```
Syslog
| where ProcessName == "sshd"
| sort by TimeGenerated desc
| take 50
```

The screenshot shows the Azure Data Explorer KQL query interface. The query editor contains the following KQL query:

```
1 Syslog
2 | where ProcessName == "sshd"
3 | sort by TimeGenerated desc
4 | take 50
5
```

The results table displays the following data:

TimeGenerated [UTC]	Computer	EventTime [UTC]	Facility	HostName	SeverityLevel	SyslogMessage	ProcessID	HostIP	Process
> 07/07/2026, 20:49:54.526	CloudSec-Linux-VM	07/07/2026, 20:49:54.000	authpriv	CloudSec-Linux-VM	info	pam_unix(sshd:session): session...	33954	172.16.0.4	sshd
> 07/07/2026, 20:49:54.518	CloudSec-Linux-VM	07/07/2026, 20:49:54.000	auth	CloudSec-Linux-VM	info	Timeout, client not responding ...	34010	172.16.0.4	sshd
> 07/07/2026, 20:48:48.160	CloudSec-Linux-VM	07/07/2026, 20:48:48.000	authpriv	CloudSec-Linux-VM	info	pam_unix(sshd:session): session...	34168	172.16.0.4	sshd
> 07/07/2026, 20:48:48.158	CloudSec-Linux-VM	07/07/2026, 20:48:48.000	auth	CloudSec-Linux-VM	info	Accepted publickey for azureus...	34168	172.16.0.4	sshd
> 07/07/2026, 20:41:53.397	CloudSec-Linux-VM	07/07/2026, 20:41:53.000	authpriv	CloudSec-Linux-VM	info	pam_unix(sshd:session): session...	33954	172.16.0.4	sshd
> 07/07/2026, 20:41:53.396	CloudSec-Linux-VM	07/07/2026, 20:41:53.000	auth	CloudSec-Linux-VM	info	Accepted publickey for azureus...	33954	172.16.0.4	sshd
> 07/07/2026, 20:41:43.716	CloudSec-Linux-VM	07/07/2026, 20:41:43.000	authpriv	CloudSec-Linux-VM	info	pam_unix(sshd:session): session...	33748	172.16.0.4	sshd
> 07/07/2026, 20:41:43.716	CloudSec-Linux-VM	07/07/2026, 20:41:43.000	auth	CloudSec-Linux-VM	info	Disconnected from user azureu...	33881	172.16.0.4	sshd
> 07/07/2026, 20:41:43.715	CloudSec-Linux-VM	07/07/2026, 20:41:43.000	auth	CloudSec-Linux-VM	info	Received disconnect from 102...	33881	172.16.0.4	sshd
> 07/07/2026, 20:40:17.341	CloudSec-Linux-VM	07/07/2026, 20:40:17.000	authpriv	CloudSec-Linux-VM	info	pam_unix(sshd:session): session...	33748	172.16.0.4	sshd
> 07/07/2026, 20:40:17.339	CloudSec-Linux-VM	07/07/2026, 20:40:17.000	auth	CloudSec-Linux-VM	info	Accepted publickey for azureus...	33748	172.16.0.4	sshd
> 07/07/2026, 20:38:26.320	CloudSec-Linux-VM	07/07/2026, 20:38:26.000	auth	CloudSec-Linux-VM	info	Connection reset by authentica...	33701	172.16.0.4	sshd
> 07/07/2026, 20:32:04.796	CloudSec-Linux-VM	07/07/2026, 20:32:04.000	auth	CloudSec-Linux-VM	info	Connection reset by authentica...	33573	172.16.0.4	sshd

Figure 9: Initial KQL query returning the most recent 50 sshd-related Syslog events.

Refinement

The query was then refined to isolate successful public-key authentication events specifically, projecting only the relevant fields for a cleaner security review.

```
Syslog
| where ProcessName == "sshd"
| where SyslogMessage contains "Accepted"
| project TimeGenerated, Computer, HostName, SyslogMessage
| sort by TimeGenerated desc
```

```

1 Syslog
2 | where ProcessName == "sshd"
3 | where SyslogMessage contains "Accepted"
4 | project TimeGenerated, Computer, HostName, SyslogMessage
5 | sort by TimeGenerated desc

```

TimeGenerated [UTC]	Computer	HostName	SyslogMessage
> 07/07/2026, 20:48:158	CloudSec-Linux-VM	CloudSec-Linux-VM	Accepted publickey for azureuser from 102.221.201.131 port 9485 ssh2: RSA SHA2564/hL56H8Nw4Kzctlin2oEzCPe03pHnOleapV+J5yZA
> 07/07/2026, 20:41:53.396	CloudSec-Linux-VM	CloudSec-Linux-VM	Accepted publickey for azureuser from 102.221.201.131 port 9470 ssh2: RSA SHA2564/hL56H8Nw4Kzctlin2oEzCPe03pHnOleapV+J5yZA
> 07/07/2026, 20:40:17.339	CloudSec-Linux-VM	CloudSec-Linux-VM	Accepted publickey for azureuser from 102.221.201.131 port 9404 ssh2: RSA SHA2564/hL56H8Nw4Kzctlin2oEzCPe03pHnOleapV+J5yZA

Figure 10: Refined query isolating successful SSH public-key authentication events.

Observation

Each result confirms a successful key-based login for the azureuser account from the trusted source IP (102.221.201.131), matching the NSG rule configured in Phase 2 and confirming the authentication method is working as designed.

6.2 Query 2 — SSH Session Activity

Purpose

This query tracked the full session lifecycle by filtering for PAM (Pluggable Authentication Module) session messages, showing when SSH sessions were opened and closed for the azureuser account.

```

Syslog
| where ProcessName == "sshd"
| where SyslogMessage contains "session"
| project TimeGenerated, SyslogMessage
| sort by TimeGenerated desc

```

```

1 Syslog
2 | where ProcessName == "sshd"
3 | where SyslogMessage contains "session"
4 | project TimeGenerated, SyslogMessage
5 | sort by TimeGenerated desc

```

TimeGenerated [UTC]	SyslogMessage
> 07/07/2026, 20:56:49.258	pam_unix(sshd:session): session closed for user azureuser
> 07/07/2026, 20:49:54.526	pam_unix(sshd:session): session closed for user azureuser
> 07/07/2026, 20:48:48.160	pam_unix(sshd:session): session opened for user azureuser(uid=1000) by azureuser(uid=0)
> 07/07/2026, 20:41:53.397	pam_unix(sshd:session): session opened for user azureuser(uid=1000) by azureuser(uid=0)
> 07/07/2026, 20:41:43.716	pam_unix(sshd:session): session closed for user azureuser
> 07/07/2026, 20:40:17.341	pam_unix(sshd:session): session opened for user azureuser(uid=1000) by azureuser(uid=0)

Figure 11: KQL query showing SSH session open and close events for the azureuser account.

Observation

The results show clear pairs of session-opened and session-closed events, giving visibility into the duration and frequency of remote administrative access to the VM.

6.3 Query 3 — SSH Disconnect Events

Purpose

This query isolated disconnect events, which can help identify abnormal session terminations, timeouts, or potential connection-based attack patterns.

Syslog

```
| where ProcessName == "sshd"
| where SyslogMessage contains "disconnect"
| project TimeGenerated, SyslogMessage
| sort by TimeGenerated desc
```

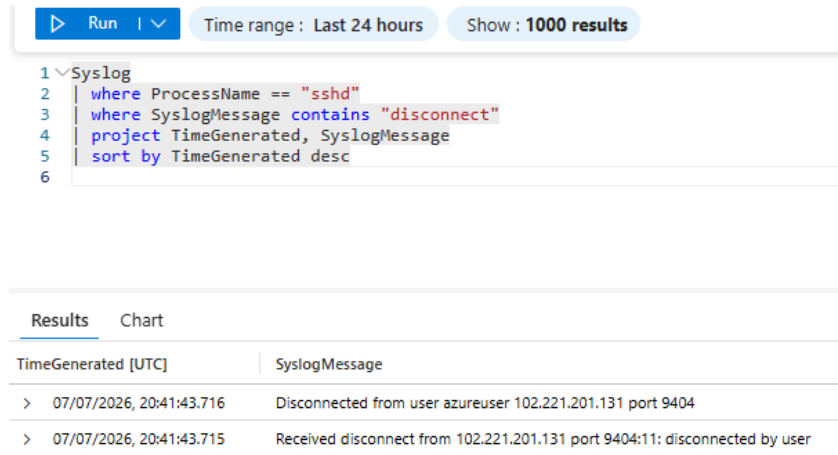


Figure 12: KQL query showing user-initiated SSH disconnect events.

Observation

The disconnect events confirm the session was closed by the user rather than terminated unexpectedly, indicating normal, expected connection behaviour for this test session.

7. Phase 6 — Automated Alerting

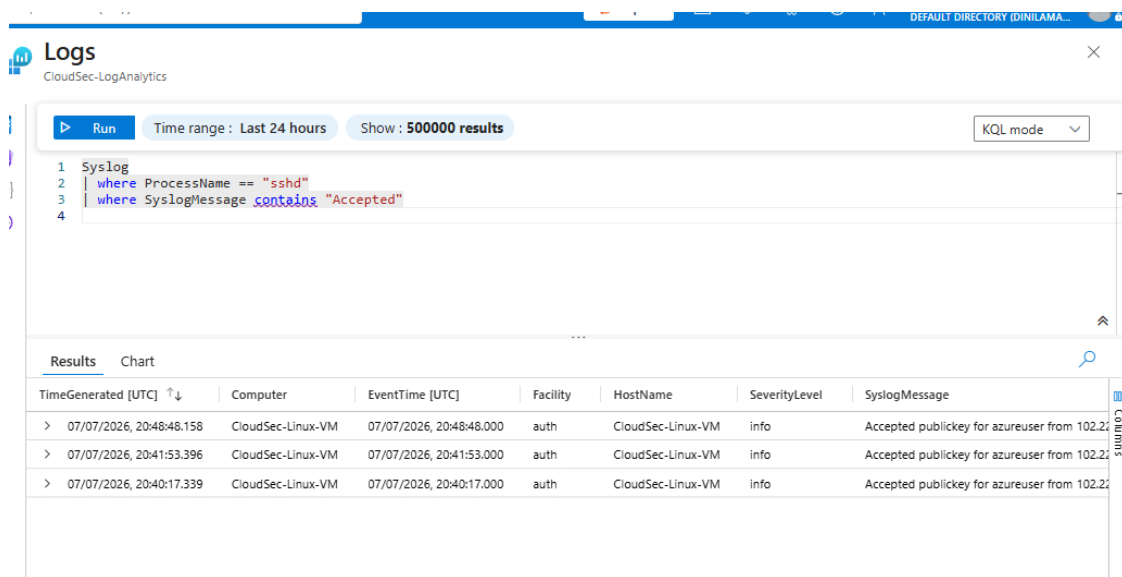
The final phase converted manual log analysis into automated, real-time detection by configuring an Azure Monitor log alert rule based on the SSH authentication query.

7.1 Alert Query Logic

Purpose

The accepted-authentication query was used as the basis for a scheduled log alert rule, so that every successful SSH login would be automatically evaluated against the alert condition rather than relying on an analyst manually running queries.

```
Syslog
| where ProcessName == "sshd"
| where SyslogMessage contains "Accepted"
```



The screenshot shows the Azure Monitor Logs interface for 'CloudSec-LogAnalytics'. A KQL query is entered in the editor: `1 Syslog`, `2 | where ProcessName == "sshd"`, `3 | where SyslogMessage contains "Accepted"`, `4`. The results are displayed in a table with columns: TimeGenerated [UTC], Computer, EventTime [UTC], Facility, HostName, SeverityLevel, and SyslogMessage. The table shows three log entries for 'CloudSec-Linux-VM' with 'Accepted' status.

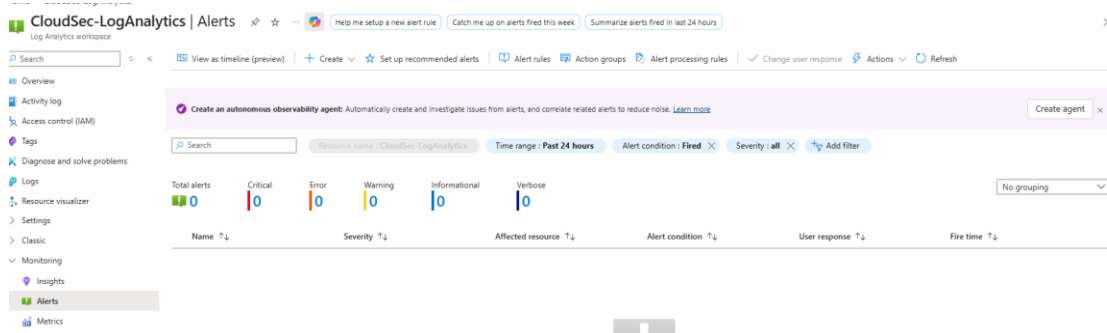
TimeGenerated [UTC]	Computer	EventTime [UTC]	Facility	HostName	SeverityLevel	SyslogMessage
07/07/2026, 20:48:48.158	CloudSec-Linux-VM	07/07/2026, 20:48:48.000	auth	CloudSec-Linux-VM	info	Accepted publickey for azureuser from 102.2
07/07/2026, 20:41:53.396	CloudSec-Linux-VM	07/07/2026, 20:41:53.000	auth	CloudSec-Linux-VM	info	Accepted publickey for azureuser from 102.2
07/07/2026, 20:40:17.339	CloudSec-Linux-VM	07/07/2026, 20:40:17.000	auth	CloudSec-Linux-VM	info	Accepted publickey for azureuser from 102.2

Figure 13: Log query used as the trigger condition for the SSH login alert rule.

7.2 Alerts Dashboard

Purpose

The Alerts blade of the Log Analytics workspace was reviewed to confirm the alert rule was correctly registered and being evaluated against incoming log data.



The screenshot shows the Azure Monitor Alerts dashboard for 'CloudSec-LogAnalytics'. It displays a summary of alerts with counts for Total alerts (0), Critical (0), Error (0), Warning (0), Informational (0), and Verbose (0). The dashboard includes filters for Time range (Past 24 hours), Alert condition (Fired), and Severity (all). A table at the bottom shows columns for Name, Severity, Affected resource, Alert condition, User response, and Fire time.

Name	Severity	Affected resource	Alert condition	User response	Fire time

Figure 14: Azure Monitor Alerts overview for CloudSec-LogAnalytics, showing the alert condition configured and monitored over the past 24 hours.

Result

The alert configuration completed the monitoring pipeline end-to-end: from raw Syslog collection, through KQL-based detection logic, to automated alerting — demonstrating a functional, minimal SOC-style detection capability built entirely on native Azure tooling.

8. Conclusion & Skills Demonstrated

This lab demonstrates practical, hands-on competency across the core pillars of cloud security operations: secure infrastructure provisioning, network access control, centralised log management, security log analysis using KQL, and automated threat detection through Azure Monitor alerting.

8.1 Key Skills Demonstrated

- Azure resource management and Infrastructure-as-a-Service (IaaS) deployment
- Network security hardening using NSGs and the principle of least privilege
- Linux server administration and SSH key-based authentication
- Centralised log management using Azure Monitor and Log Analytics
- Data Collection Rule (DCR) configuration for Syslog ingestion
- Security log analysis and threat hunting using Kusto Query Language (KQL)
- Configuration of automated, alert-based security monitoring

8.2 Potential Next Steps

- Integrate Microsoft Sentinel for SIEM-level correlation and incident response workflows.
- Extend the alert rule with an Action Group to trigger email/SMS notifications or automated remediation (e.g. via Logic Apps).
- Simulate brute-force SSH attacks to validate alert sensitivity and tune detection thresholds.
- Apply Microsoft Entra Conditional Access policies to further restrict administrative sign-in.